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Question Paper Code : 21585

B.E. / B.TECH. DEGREE EXAMINATIONS, NOVEMBER / DECEMBER 2024

Third Semester

Computer Science and Engineering

23CS304 – PROGRAMMING WITH JAVA

(Regulations: Mepco – R2023)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL,CO **PART A – (10 × 2 = 20 Marks)**

U,CO1 1. Write about constructors and their use.

A,CO1 2. What is the output of the following code fragment? Justify your answer.

```
int [ ] odd = {1, 3, 5, 7, 9, 11 };
System.out.println( odd[0] + "  " + odd[30] ) ;
```

A) 1 5 B)1 -1 C) 1 7 D) error

U,CO2 3. Explain the different usage of final keyword.

U,CO2 4. We have created a class named Vehicle. Another class named TwoWheeler is inherited from Vehicle. Class FourWheeler is inherited from Vehicle. Class Bicycle and Class Bike are inherited from TwoWheeler. What type of inheritance is this? Justify your answer.

A) Multiple inheritance B) Multilevel inheritance

C) Single Level inheritance D) Hierarchical inheritance

U,CO3 5. Differentiate AWT framework with Swing framework.

U,CO3 6. Write about the methods used to execute SQL SELECT query using Java JDBC.

A,CO4 7. Consider the following Java code snippet that uses generics. Identify the output / error if any. Justify your answer.

```
class Container<T> {
    private T item;
    public void setItem(T item) {
        this.item = item;
    }
    public T getItem() {
        return item;
    }
}
public class GenericsErrorTest {
    public static void main(String[] args) {
        Container<String> stringContainer = new Container<>();
        stringContainer.setItem("Hello");
        Container<Integer> integerContainer = new Container<>();
```

```
        integerContainer.setItem(100);
        String item = integerContainer.getItem();
        System.out.println(item);
    }
}
```

A) The code compiles successfully and prints 100.
B) The code compiles successfully and prints Hello.
C) The code does not compile due to a type mismatch error.
D) The code throws a runtime error due to an incorrect type cast.

A,CO4 8. Identify the output of the following code fragment. Justify your answer.

```
import java.util.*;
public class CollectionsTest {
    public static void main(String[] args) {
        Set<String> set = new LinkedHashSet<>();
        set.add("Apple");
        set.add("Banana");
        set.add("Apple");
        set.add("Cherry");
        System.out.println(set);
    }
}
```

A) Apple, Banana, Apple, Cherry B) Apple, Banana, Cherry
C) Apple, Cherry, Banana D) Banana, Apple, Cherry

U,CO5 9. Write short notes on thread priorities and how to set them.

A,CO5 10. Which method is used to make the current thread give its execution and allow other threads of the same priority to execute? Justify your answer.

A) Thread.sleep() B) Thread.join()
C) Thread.yield() D) Thread.interrupt()

PART B – (4 × 15 = 60 Marks)

A, CO1 11. a) Develop the Java code for the following application.

Create a class named "Person" which has name, age, aadharnumber as data members and getInput( ), display( ) and licenceApproval( ) as member funtions.

Handle the following Exceptions:

A) If the age has alphabets then throw NumberFormatException
(Hint: Throw built-in exception)

B) If Aadhar number is not having 12-digit numeric value then throw InvalidAadharNumberException
(Hint: Throw user-defined exception)

C) For getting licence if age is less than 18 and the person is not having aadhar then throw NotEligibleForLicence
(Hint: Throw user-defined exception) (15 Marks)

OR

A, CO1 11. b) Create a package named “converter” with a class “TemparatureConverter” with functions to convert Fahrenheit to Celsius and vice-versa. Create another class called “DistanceConverter” with functions to convert KM to M and viceversa. Develop a Java program to illustrate the conversion of temperature and distance using a menu driven approach. The functions should be invoked without creating objects for the package classes.
[F° to C°: subtract 32 and multiply by 5/9]
[C° to F°: multiply by 9/5 and add 32] (15 Marks)

A, CO2 12. a) i. Consider the following Java skeleton:
Class B extends A implements C { ….. }

A) Analyze the components A, B, C and develop code for A and C.
B) Identify the invalid statements in the following:

Justify your answer.

```
A a = new A( );
B b = new B( );
C c = new C( );
A aa = new B( );
B bb = new C( );
C cc = new B( );
B bbb = new A( );
```

C) Explain the type of inheritance in this skeleton. (10 Marks)

U, CO2 12. a) ii. Write short notes on reflections with an example. (5 Marks)

OR

A, CO2 12. b) i. Create a zoo management system in Java using inheritance and interfaces. Create an abstract class Animal with an abstract method makeSound() and a concrete method eat(). Implement an interface Habitat with methods getHabitat() and setHabitat(). Define concrete classes Lion and Parrot that extend Animal and implement Habitat, and write a main method to demonstrate their functionality. (10 Marks)

U, CO2 12. b) ii. Compare the advantages of using an abstract class versus an interface in the above context. Assess how the design allows for extensibility and code reuse in managing different types of animals and their habitats. (5 Marks)

A, CO4 13. a) Create a generic class *MultiValueMap<K, V>* in Java that allows storing multiple values for a single key. The class should support the following operations:

A) Add a key-value pair.
B) Retrieve all values for a specific key.

C) Remove a specific value associated with a key or remove the key and all its values.
D) Check if a key or value exists in the map.
E) Display all key-value pairs.

Develop a *main method* to demonstrate its functionality with various data types and scenarios. (15 Marks)

OR

A, CO4 13. b) Create a generic class *GenericLinkedList<T>* in Java that represents a singly linked list. The class should support the following operations:

A) Add an element to the end of the list.
B) Remove the first occurrence of a specified element.
C) Retrieve the element at a specified position (index).
D) Display all elements in the list.
E) Count the occurrences of a particular element.

Implement the *GenericLinkedList class* using a nested *Node* class to represent individual elements of the list. Provide a `main` method to demonstrate its functionality with different data types (e.g., *Integer, String)* by performing the add, remove, retrieve, and display operations. (15 Marks)

A, CO5 14. a) i. Develop a Java program with two threads: one *(IncrementThread)* that increments a shared counter every second for 10 seconds, and another *(PrintThread)* that prints the counter value every 2 seconds for 10 seconds. Ensure thread-safe operations on the shared counter using synchronization mechanisms. Provide a *main method* to start and manage both threads. (10 Marks)

U, CO5 14. a) ii. Write short notes on thread states and its life cycle. (5 Marks)

OR

A, CO5 14.b) i. Develop a Java program with two threads: one *(ProducerThread*) that generates and adds random integers to a shared buffer every second for 10 seconds, and another *(ConsumerThread)* that removes and prints integers from the buffer every 2 seconds for 10 seconds. Ensure thread safety and synchronization for access to the shared buffer. Implement a *main method* to start and coordinate both threads. (10 Marks)

U, CO5 14. b) ii. Briefly explain about the thread synchronization in inter thread communication. (5 Marks)

PART C – (1 × 20 = 20 Marks)

A, CO3 15. a) Develop a Java Swing application for an airport check-in system. The application should include the necessary components to get the inputs from the user. The check-in system should have the following features:

A) Validation Rules:
   - Passenger Name: Must be at least 3 characters long and start with a letter.
   - Passport Number: Must be exactly 9 alphanumeric characters.
   - Flight Number: Must be exactly 6 alphanumeric characters.
   - Check-in Date: Must be today's date or a future date.
   - Agree to Terms and Conditions: Must be checked to complete the check-in process.

B) Functionality:
   - Display the check-in details in a designated area on the right side of the form upon successful check-in, along with a confirmation message at the bottom.
   - Clear all fields and display a "Check-in Unsuccessful" message if any validation errors occur. (20 Marks)

OR

A, CO3 15. b) Design and implement a Java Swing application for a library book return system with necessary components and with the following requirements:

A) Validation Rules:
   - Member ID: Must be exactly 6 alphanumeric characters.
   - Book ISBN: Must be exactly 13 digits long. Book ISBN should be already issued to the member. if not show an error message.
   - Issue Date: Must be a valid date and cannot be a future date.
   - Return Date: Must be a valid date and should not be earlier than the issue date.
   - Fine Calculation: Calculate a fine of Rs.1 per day if the book is returned late.

B) Functionality:
   - Display the book return details and any applicable fines in a designated area on the right side of the form upon successful return.
   - Clear all fields and display a "Return Unsuccessful" message if any validation fails or errors occur. (20 Marks)